

BOWLING GREEN, KY, USA

WMO: 746716

Lat: 36.965N

Lon: 86.424W

Elev: 161

StdP: 99.41

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93808

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.4	(c) -8.4	(d) -17.5	(e) 0.8	(f) -8.8	(g) -14.8	(h) 1.1	(i) -7.0	(j) 10.5	(k) 13.8	(l) 9.2	(m) 11.2	(n) 2.2	(o) 330	(p) 0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.5	24.0	33.2	24.0	32.2	23.7	25.8	31.6	25.3	30.9	24.7	30.0	3.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.4	28.9	23.6	18.8	28.2	23.0	18.1	27.6	80.5	31.8	78.0	31.0	75.5	30.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.9	(b) 7.8	(c) 7.0	(e) -14.6	(f) 36.9	(g) 3.2	(h) 2.6	(i) -16.8	(j) 38.8	(k) -18.7	(l) 40.3	(m) -20.5	(n) 41.8	(o) -22.8	(p) 43.7		
(a) 8.9	(b) 7.8	(c) 7.0	(e) -15.2	(f) 27.0	(g) 3.0	(h) 0.9	(i) -17.4	(j) 27.7	(k) -19.1	(l) 28.2	(m) -20.8	(n) 28.7	(o) -23.0	(p) 29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	2.4	4.8	9.2	14.8	19.7	24.1	26.0	25.4	21.9	15.4	8.9	4.5
	DBStd	9.54	6.63	6.01	5.73	4.99	4.12	2.90	2.47	2.57	3.73	4.75	5.08	5.68
	HDD10.0	790	251	165	84	14	1	0	0	0	0	9	78	187
	HDD18.3	2162	495	378	286	127	36	2	0	0	12	116	283	428
	CDD10.0	2544	14	20	61	158	301	423	495	477	356	176	47	18
	CDD18.3	878	0	1	4	21	78	174	237	219	118	24	2	1
	CDH23.3	8651	0	2	34	190	708	1701	2447	2167	1128	259	14	0
	CDH26.7	3424	0	0	3	30	203	679	1063	933	448	63	2	0
Wind	WSAvg	2.7	3.2	3.3	3.3	3.4	2.8	2.4	2.2	2.0	1.9	2.2	2.6	2.9
Precipitation	PrecAvg	1269	94	103	124	106	126	105	109	92	94	77	109	120
	PrecMax	1945	264	286	370	246	348	282	227	212	298	196	214	334
	PrecMin	796	23	18	42	24	35	27	30	18	7	2	18	27
	PrecStd	218	54	55	65	53	68	60	54	49	60	45	47	66
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.3	22.8	26.6	29.2	31.9	34.3	36.6	36.8	35.5	30.7	25.3	21.2
		MCWB	16.5	16.5	18.0	19.6	22.7	23.8	25.5	24.1	22.1	21.4	18.4	17.0
	2%	DB	18.0	19.9	23.9	27.4	30.3	33.0	34.1	34.4	33.1	28.4	22.4	18.4
		MCWB	15.0	15.0	16.4	18.7	21.7	23.8	25.0	24.0	22.4	19.8	16.3	15.0
	5%	DB	15.9	17.6	21.4	25.7	29.0	32.0	32.9	32.8	31.3	26.3	20.1	16.1
		MCWB	13.2	13.2	14.9	17.9	21.0	23.5	24.7	23.8	22.2	18.7	15.1	13.2
	10%	DB	13.2	15.0	19.0	23.7	27.5	30.7	31.7	31.4	29.2	24.1	17.8	13.8
		MCWB	10.3	11.4	13.4	16.8	20.7	22.9	24.1	23.5	21.3	18.1	13.5	11.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	17.9	19.1	21.6	24.3	25.9	26.9	26.4	24.8	23.5	19.6	17.9
		MCDB	19.2	21.4	24.0	27.1	29.3	31.6	33.0	33.1	30.8	28.0	22.9	20.1
	2%	WB	15.6	16.2	17.6	20.3	23.1	25.2	26.2	25.5	24.0	21.8	17.9	15.6
		MCDB	17.6	19.2	22.8	25.4	28.4	31.0	32.0	31.4	29.8	26.1	21.2	17.9
	5%	WB	13.7	14.1	16.0	19.1	22.2	24.5	25.6	24.9	23.3	20.3	16.0	13.6
		MCDB	15.7	16.7	20.4	23.6	27.4	30.0	31.2	30.4	28.6	24.4	19.2	15.7
	10%	WB	10.6	11.6	14.3	17.9	21.3	23.8	25.0	24.3	22.5	19.0	14.1	11.4
		MCDB	12.8	14.4	18.3	22.4	26.1	28.9	30.3	29.5	27.4	23.0	16.9	13.6
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.2	11.4	12.2	11.2	10.7	10.3	11.0	12.1	12.6	11.4	9.3
		MCDBR	11.3	13.5	14.6	14.0	12.8	12.3	11.6	12.5	14.3	14.9	14.5	11.7
		MCWBR	8.7	9.6	8.8	7.4	5.8	4.8	4.2	4.5	5.3	7.4	9.3	9.1
	5% WB	MCDBR	10.5	11.6	12.4	11.4	10.9	10.5	10.1	10.5	11.2	11.8	11.5	9.9
		MCWBR	9.0	9.4	8.3	7.0	5.6	4.7	4.3	4.3	4.9	6.7	8.6	8.8
Clear-Sky Solar Irradiance	taub	0.317	0.325	0.369	0.407	0.450	0.472	0.506	0.517	0.427	0.369	0.333	0.310	
	taud	2.462	2.423	2.346	2.243	2.168	2.141	2.065	2.018	2.256	2.404	2.464	2.489	
	Ebn at Noon	872	909	894	874	839	818	786	766	823	846	845	856	
	Edh at Noon	84	98	116	136	149	153	164	169	126	99	83	76	
All-Sky Solar Radiation	RadAvg	2.08	2.70	3.79	5.02	5.61	6.24	5.95	5.52	4.63	3.53	2.49	1.79	
	RadStd	0.20	0.35	0.36	0.35	0.50	0.37	0.35	0.29	0.40	0.41	0.26	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.84	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+131
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+123	+80

Nomenclature: See separate page